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Computing and testing extensive total viewsheds: a case of prehistoric burial mounds in Bohemia

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A B S T R A C T

This paper reports on the progress in the field of total viewshed computation, which was made by the joint efforts of several Slovak and Czech research institutions. The result of the project is eight total viewsheds calculated for the whole territory of Bohemia (57,000 km2) with a relatively high resolution (grid from 5 × 5 to 320 × 320 m, visibility radii from 0.5 to 64 km). Through the principal component analysis, the viewsheds were further processed and derived datasets were created defining the most important gradients of visual affordances of the Bohemian landscape. On a sample of archaeological sites (barrow cemeteries from a part of South Bohemia), it was examined how the resulting models could be interpreted and whether the identified visibility gradients affected people's behaviour in the past, namely the location of their funerary monuments. On the one hand, this study shows that non-random relationships of this type existed, but on the other hand, visibility itself may not always have been the only (or the main) factor in past behaviour, even if it is related to above-standard places in terms of visibility. All total-viewshed datasets are made available for further research through public repository.

1. Introduction

The main aim of this paper is to report on the successful creation of total-visibility models for a large territory and to indicate some new possibilities of their use in landscape archaeology. The models were created for the whole territory of Bohemia (incl. the Vysočina Region), i.e., for the western part of the Czech Republic, covering an area of more than 57,000 km2 in a relatively high resolution (Fig. 1). They were built in the course of international collaboration between the Institute of Archaeology of the CAS in Prague and the Slovak University of Technology in Bratislava, where the new modelling method for total viewshed was developed and tested on the territory of Slovakia. Some more institutions took part in the preparation of the data for its first application.[1]

Already in the 1990s, the Institute of Archaeology in Prague experimented with total visibility modelling, but at that time, it was impossible to obtain sufficient computing capacities to handle the task. Today, the situation is changing, but for most archaeological institutions the problem of computational processing is still hard to overcome. As a result, attempts at the use of total viewsheds in archaeology are usually applied only on relatively small territories (Llobera et al., 2010; Brughmans et al., 2018). The goal of our project was (i) to find an approach that would allow breaking the given capacity limits, (ii) with the help of which we would be able to create total viewsheds of the whole territory of Bohemia and to discuss their contents and relevance, (iii) to verify the usability of total-visibility models on a specific archaeological example and (iv) to provide our viewshed data for the public to be applied in various fields of research and other spheres of potential use.

The results of the project have shown that the research use of total viewsheds in landscape studies raises a broad range of other theoretical and technical questions. Among them, e.g., the concept of 'place' as a coherent and meaningful unit (contrasting to a group of cells in a raster dataset) could be mentioned, recalling some approaches of landscape archaeology (e.g., Tilley 1994; Bradley 2000; Wheatley 2004; Bowser - Zedeño eds. 2009 and others). This text presents basic information about

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our visibility models and attempts to use them in archaeology, using a set of prehistoric barrow cemeteries in one of the regions of South Bohemia.

2. Background

By landscape we understand an environment shaped and perceived by people (Council of Europe Landscape Convention, Article 1); the visibility of its surface is considered one of its key features. Although it is justified to stress that visual perception is only one way for humans to orient themselves in the landscape (Wheatley – Gillings 2000; Frieman – Gillings 2007; Llobera 2007; Gillings 2017), for many reasons, it seems to be the most important way. Intentionally or unintentionally, people structure the landscape and their activities in space in such a way as to make its visual aspects correspond to their specific needs – from practical regards (control of the territory and its protection) to symbolic meanings (e.g., organising space according to visible ideological centres). It is of course impossible to study perceptions of people in the past directly, but we can assume that their effects were projected to cultural patterns that are reflected in the material world and therefore – to a certain extent – contained in the archaeological context. This relation can be seen as *vice versa* as well: the landscape has certain visual properties (affordances, Gibson 1986), which may be used by every society differently. If we can identify relevant visual properties of the landscape environment, we are also able to explore whether and to what degree they are connected with patterns in the archaeological context, to estimate which meanings they were ascribed by past communities and what this relationship reveals about the character of a given past society.

Some decades ago, geographic information systems (GIS) provided a completely new tool for modelling the properties of landscape (Lock – Stančić eds. 1995; Lake et al. 1998; Lake – Woodman 2003; Jacobson 2007). Visibility analysis is mainly concerned with the modelling of *viewsheds* that allows describing a place in the landscape by size and position of the area in sight (field-of-view, Gillings – Wheatley 2020). Archaeology is then able to deduce whether good visibility (or hiddenness) played a role in the choice of the activity area and whether it is in some way related to the sort of activities conducted on the site (Kuzma – Lieskovský 2007). Based on this principle, we are also able to create chains of inter-visible places and to find out if their visual interconnection had any significance (Llobera 2006; 2007). A combination of more viewsheds (current technology allows creating and connecting viewsheds even of thousands of points: Gillings 2017) is denoted *cumulative viewshed* (Wheatley 1995).

In principle, both the simple and cumulative viewsheds describe the landscape from the viewpoint of the observer. We know or assume where people stood and study what they could have seen from there; we can also compare various sets of real and/or hypothetical sites regarding their viewshed areas. This method offers many advantages for the interpretation of the visual relations, but it also contains a certain disadvantage: we do not know about the qualities of the surrounding landscape and the relative position of our site(s) on the scale of general visual affordances. The calculation of the *total viewshed* (cf. Llobera et al. 2010), which is the subject of this article, solves this problem. Total viewsheds are not dependent on any specific observer but calculate the visibility of each point in the landscape from all other points within the selected radius.

Here, the roles of the observer and the target are reversed: we exactly know what is viewed (a given point in the landscape), but we do not know from where, or we rather suppose that from everywhere. The total visibility is here understood as an important attribute of the landscape itself, one of its affordances (Gibson 1986; Gillings 2009; Wernke et al. 2017; Gillings – Wheatley 2020). Removing a specific observer is also connected with some disadvantages since it is improbable that all potential viewpoints had the same significance for the observer; in the future, we should find a way to scale the input values more precisely (cf. *affordance viewshed*: Gillings 2009; 2012).

In our project, we have processed eight total viewsheds for the whole of Bohemia. These viewsheds differ in their supposed visibility radii: 0.5,

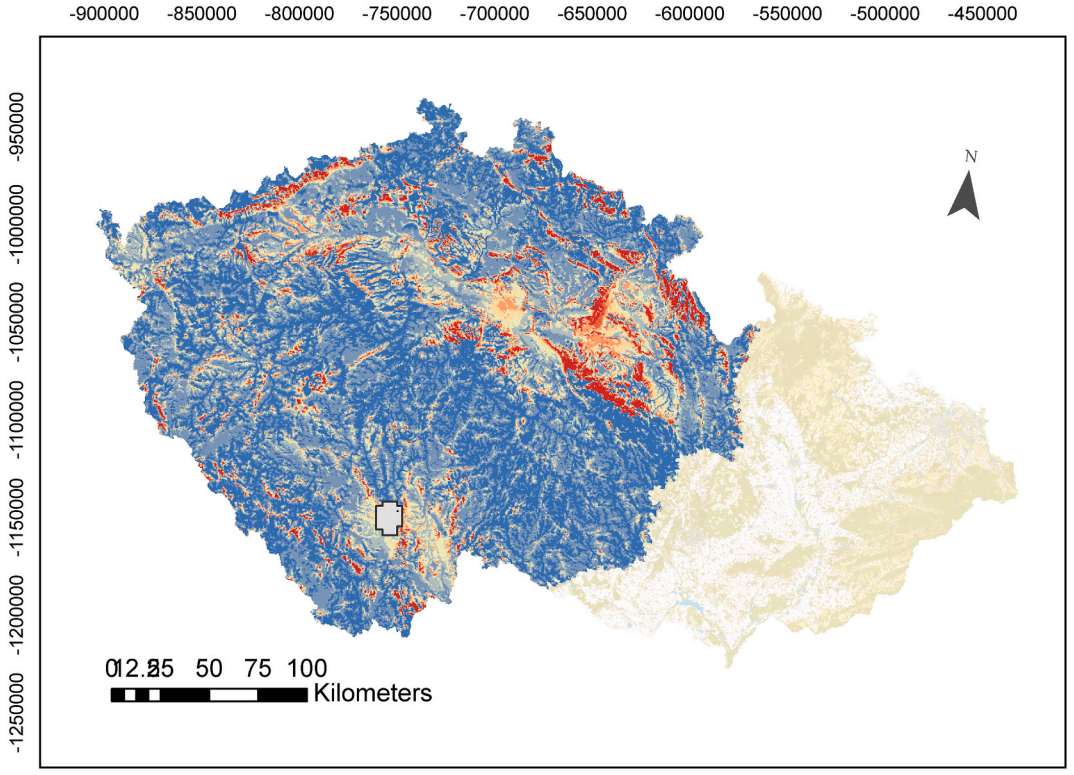


Fig. 1. Map of the Czech Republic showing the extent of the total viewsheds (Bohemia) and the area of detailed study around Hluboká nad Vltavou, South Bohemia. Displayed in the S-JTSK/Křovák East North grid. Figures by M. Kuna (unless otherwise stated).

1, 2, 4, 8, 16, 32 and 64 km (see 3.1 for details). We also made them available to the public for reuse by researchers and other interested parties. Probably each of these models can be used only for a certain set of research questions and under certain conditions. While the short-distance viewsheds can be related to individual settlement features, the everyday economic strategies of prehistoric communities etc., long-distance visibility could have played a role, e.g., in founding regional centres. In the latter case, individual details and man-made structures did not need to – and often could not – have been visible, contrary to larger natural landforms (e.g., hills), with which the human activities in social consciousness may have been related.

Empirical archaeological data, which is here analysed to demonstrate the relevance of total viewsheds, consists of prehistoric barrow cemeteries in the area of Hluboká nad Vltavou, South Bohemia (Figs. 1 and 8:B–C). The given set of archaeological sites is unique within Bohemia and perhaps even in a wider context. Prehistoric burial mounds (from the Bronze and Hallstatt Ages, c. 2000–400 BCE) have been preserved there in a high number (119 cemeteries with 1039 burial mounds on a territory of 196 km2) mainly due to the population displacement in the early modern times (16th century) and the transformation of the whole territory into game reserves. The Bronze Age is generally associated with the beginnings of regular settlement in southern Bohemia, which is characterised by a somewhat harsher climate, but good connections with the Danube region and access to various raw materials (e.g., copper from the Alps). Although the settlement character probably did not differ much from other parts of Bohemia, due to the combination of specific soil conditions and modern land use, its archaeological manifestation is specific (settlements are usually neither abundant nor rich, but a large number of mounds have been preserved). Most likely, in the Bronze Age to Hallstatt periods, a network of smaller settlements was created and accompanied by burial grounds, which often show long-term continuity (Beneš et al. 1999; Hlásek – Chvojka 2020).

Location analysis of the barrows has already been the subject of one of our previous studies (Kuna 2006; 2008); in this paper, we try to extend this research to the issue of visibility and to use this primarily as a demonstration of the problems that arise when using total viewsheds as an analytical input. The assumption that the location of the prehistoric burial mounds may generally be connected with their good visibility seems justified considering their frequent position on hilltops, upper slopes, ridges and other types of dominant and well-visible places in the landscape (Kasbohm 1955; Torbrügge 1959; Kristiansen 1998; Sahlqvist 2000; Kuna 2008).

3. Method[2]

3.1. Calculation of total viewsheds and their basic properties

The LiDAR-based *Digital Terrain Model of the Czech Republic of the* 5th *Generation* (DMR 5G) was used as input for the calculations.[3] This bare-earth model with a spatial resolution of 2 × 2 m and a vertical accuracy of 0.18–0.3 m was derived from the data of aerial laser scanning in 2010–13. Before the calculation, we removed most modern interventions present in the model, which would have considerably changed the visual properties of the natural relief. Modern landscape elements (e.g., embankments of railways and roads, quarries, etc.) were identified in topographic maps and cut out; their surface was then interpolated (for details see Novák – Pružinec 2022). The territory under study was enlarged by the observing radius to avoid edge effect: MERIT DEM (Yamazaki et al. 2017) was used for the territory outside DMR 5G to create a rectangular raster leyer used as the input DEM in the computation.

In practice, the calculation of total viewsheds is limited by the immense computational time and hardware requirements. Various approaches have been published to optimise this task. Some authors employed parallelization (Llobera et al. 2010; Mineter et al. 2003; Jakab – Petluš 2012), others focused on GPU utilisation (e.g. Strnad 2011; Zhao et al. 2013) or algorithm modification (Tabik et al. 2013, 2015; Yu et al. 2017; Rana 2003). However, even with large grids and optimisations, the total-viewshed computation may still be too demanding without adequate computational resources: the hardware and time requirements may soon exceed the capacities of conventional computers. Therefore, we employed both parallel computation and easy-to-implement modifications of the viewshed algorithm in MATLAB, which is commonly supported in high-performance computing clusters.

In 2019, the capacity of 400,000 computing hours was provided to our project by the IT4Innovations research infrastructure within its supercomputer. Finally, it took about 256,000 CPU hours to compute the resulting eight total-visibility models for the selected territory (Table 2; additional computing capacity was used for related testing and optimising of the calculations and input data). These models are now freely available for research and other types of use (Bucha Rášová et al. 2021), allowing to avoid recurring costs in the form of immense computational resources.

Our viewshed calculation is based on the R2 algorithm in spherical approximation (see, e.g., Franklin – Ray 1994) and relies on the viewshed function from MATLAB's Mapping Toolbox. To save computational time, the function was significantly optimised by the authors for large-scale parallel computations. While achieving almost a fifty-fold speed-up factor, the outputs remained the same as those from the original function. Given that the single viewshed is called many times, these optimisations may add up and thus save a considerable amount of time. Our procedure involved optimisations as follows:

- Bilinear interpolation of DEM was done by an in-house function that is faster than the original Matlab's *interp2* function. Interpolation is needed to compute lines of sight in individual single viewsheds.
- To avoid unnecessary computations, all sub-functions were modified to compute only the values that we needed. For instance, the *greatcircleinv* sub-function computes the great circle distance and the azimuth between points on a sphere, but only the former calculation is needed to compute total viewsheds.
- Almost all sub-functions that are called multiple times were embedded into the *viewshed* function to eliminate the overhead.
- Functions computing intrinsic coordinates (e.g., *intrinsicToGeographic*) were replaced by our own faster ones that produce the same outputs.
- To determine whether a cell is visible or invisible to the observer, the time-consuming *atan2* function can be avoided, as it is possible to work directly with the tangent of the observing angle.
- The observing radius was implemented into the *viewshed* function, a feature missing in the original routine.
- The code was parallelized by calling the *single_viewshed* function inside a *parfor* loop, which is an ordinary *for* loop running in parallel on multiple threads. This enabled us to employ multiple threads available on the supercomputer.[4]

By default, each cell of the resulting total viewshed represents the number of times the cell is visible (Fig. 2). This type of raster dataset is sometimes referred to as the *dominance viewgrid* (Lee – Stucky 1998; Llobera 2003) or the 'times-seen map' (Gillings 2009; Brughmans et al. 2018; Mills et al. 1992). The large size of the territory, the high resolution of the terrain model and the resulting enormous computational complexity of the intended task led us to sample the observing points

[2] See also Table 1 summarising the steps taken and results produced.
[3] https://ags.cuzk.cz/arcgis2/rest/services/dmr5g/ImageServer.
[4] Due to the licence restrictions associated with the *viewshed* function and its sub-functions, we are not allowed by the MathWorks, Inc. to release the source code of the modified *viewshed* function.

Table 1
Description of basic workflow and a list of the main GIS raster datasets (total viewsheds and their derivatives) discussed in this paper.

No.	Step	Method	Raster dataset	Dataset description	Extent	Values
1	Acquisition of LiDAR data	Downloading DMR 5G from the Czech Cadastral Office	DEM	LiDAR-based digital elevation model	Czech Republic	m MSL
2	Filtering out modern features	Novák – Pružinec 2022	DEM_filter	DEM without most of the modern terrain features	Bohemia	m MSL
3	Transformation from projected to geographic coordinate system and extension by MERIT DEM to avoid edge effect	Yamazaki et al. 2017	DEMn-ext	7 DEMs (Bohemia with 0.5, 1, 2, 4, 8, 16 and 70 km buffer zones) extended to a rectangular shape (Matlab requirement)	rectangular extensions of Bohemia	m MSL
4	Creating cliping/sampling masks to be used in computations	Rášová 2017	DEMn-mask	7 masks for computation of the single viewsheds removing cells outside the observing radius and sampling of observing points (see Fig. 3)	buffer-extended Bohemia	0–1
5	Parallel computation of single viewsheds and combining them	Rášová 2017	TVIEWn-abs	Rasters with absolute numbers of visible cells in the given radius.	Bohemia	integer (no. of cells)
6	Transformation to projected coordinate system and calculation of total viewsheds as percentage of visible cells	Rášová 2017	TVIEWn	TVIEW1: radius - 0.5 km; cellsize = 5 m; TVIEW2: radius - 1 km; cellsize = 10 m; TVIEW3: radius - 2 km; cellsize = 20 m; TVIEW4: radius - 4 km; cellsize = 40 m; TVIEW5: radius - 8 km; cellsize = 80 m; TVIEW6: radius - 16 km; cellsize = 160 m; TVIEW7: radius - 32 km; cellsize = 320 m; TVIEW8: radius - 64 km; cellsize = 320 m	Bohemia	0–100%
7	Resampling to uniform cellsize	ESRI ArcMap Resample tool	TVIEWn-8m	TVIEWn with uniform cellsize of 8 × 8 m	Bohemia	0–100%
8	Calculating differential viewsheds	ESRI ArcMap Raster Calculator tool, using formula: ((TVIEWn * 4) – TVIEW(n–1))/3	DVIEWn	Differential total viewsheds of the nth order	Bohemia	0–100%
9	Calculating PCA	ESRI ArcMap Principal Components tool	PCAn	PCA1: visibility gradient (component) from the short and middle distances (500 m - 4 km); PCA2: gradient (component) of the contrast between the short-range visibility (500 m) and middle-range visibility (2–4 km) PCA3: visibility gradient (component) from the long distances (4/8–16/32 km)	Bohemia	0–255
10	Cropping PCA rasters to the area of interest	ESRI ArcMap Extract by Mask tool	PCAn-reg	PCAn models cropped by the area under study	Hluboká study area	0–255
11	Cropping PCA rasters by the catchment area	ESRI ArcMap Extract by Mask tool	PCAn-1km	PCAn models cropped by the buffer of 1 km around the barrow cemeteries	1 km buffer around barrows	0–255
12	Calculating mean values and standard deviations for the PCA rasters	ESRI ArcMap Focal Statistics tool	PCAn_mean/ PCAn_std	Rasters of mean values and standard deviations of PCAn in the 250 m radius	Hluboká study area	real numbers
13	Calculating relative affordances based on the PCA rasters	ESRI ArcMap Raster Calculator tool, using formula: (PCAn – PCAn_mean)/PCAn_std	RPCAn	Relative visibility gradients PCA1-3	Hluboká study area	real numbers

Table 2
Computing time achieved for the calculation of total viewsheds. *DEM size* is the number of cells in rows and columns in a rectangular DEM, *total number of cells* is the number for the area of Bohemia enlarged by the observing range; about 1/8 of these cells were used as observing points.

radius/cell size	0.5 km/5 m	1 km/10 m	2 km/20 m	4 km/40 m	8 km/80 m	16 km/160 m	32 km/320 m	64 km/320 m
No. of tiles	64	16	4	1	1	1	1	1
DEM size (rows x columns)	46727 × 90130	23481 × 45247	11825 × 22754	6000 × 11514	3092 × 5900	1630 × 3080	1092 × 1972	1092 × 1972
Total number of cells	2,541,746,181	646,541,284	165,312,416	43,144,288	11,681,255	3,313,594	1,538,469	1,538,469
CPU time (hrs)	194136.6	46318.6	12367.5	2887	232.7	27.6	8.8	17.9

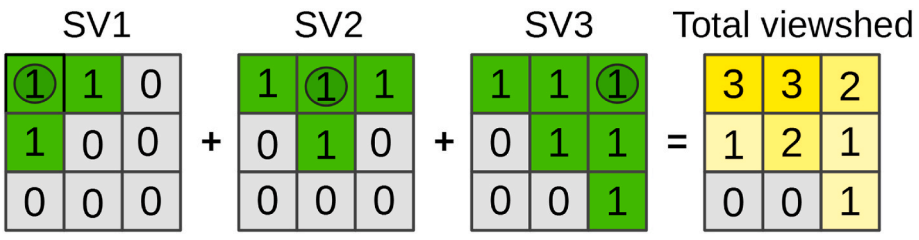


Fig. 2. Principle of the total-viewshed calculation. Single viewsheds (SV) are computed from each grid cell (here for the sake of brevity only from the upper row) and then are added up. The observation cells of single viewsheds are marked by circles. Note that the input DEM is not shown and the single viewsheds are illustrative. By A. Bucha Rášová.

when calculating the viewsheds. Specifically, we calculated single viewsheds only from every eighth point according to the scheme shown in Fig. 3. The validity of this procedure and its results were confirmed by our experiments proving that evenly distributed observing points accurately approximate the original (full) total viewshed (Rášová 2017).[5] Due to this reduction, the absolute total-viewshed values decreased as they were derived from fewer observing points; the relative values (percentages of visible cells in the given radius) were, however, mostly unaffected.

We believe that it is still correct to label our results as total (not cumulative) viewsheds, although we did not use all DEM cells as observing points, for two reasons: (i) our algorithm approximates the total viewshed and represents the same phenomenon; (ii) the term cumulative viewshed refers to a viewshed raster computed from a group of specific (selected) observing points, not a set of points evenly distributed over the landscape.

Both the observer and the target heights were set at 1.5 m, which slightly overvalues the total visibility, and at the same time, lowers the influence of deviations and the distortion of the local relief. Different settings of the observer and target height parameters were tested during the preparation phase of our computations. In principle, the smaller the value is set, the more the local micro-relief captured by LiDAR influences the result. In the case of using zero values, the influence of microrelief is very high and even the smallest landform can strongly affect the total visibility values, especially in the case of terrain ridges where each cell can be easily hidden behind the horizon (see Fig. 4). Since the goal of the model creation was to obtain proxy data for studying past landscapes whose exact micro-relief is unknown, we used a value of 1.5 m for the height of the observer and the target identically, which roughly corresponds to a standing person. This arbitrary value essentially means that terrain variations up to 1.5 m should not affect the results, the impact from analytically marginal terrain changes is suppressed and the resulting values are desirably spatially homogenised.

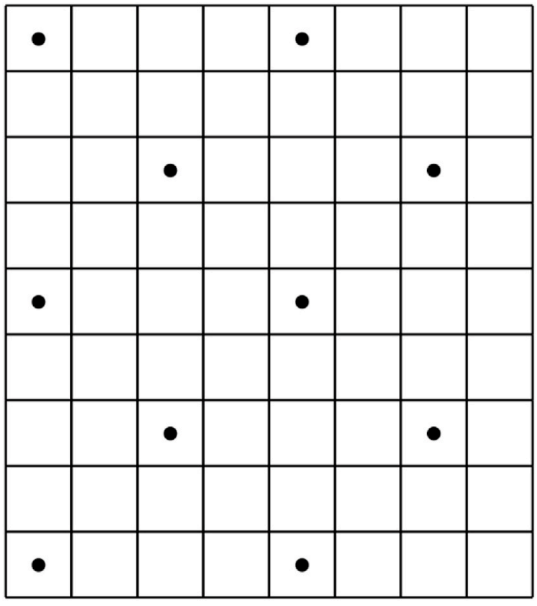


Fig. 3. Scheme of the total-viewshed calculation. A total viewshed is created as a sum of single viewsheds calculated from only the selected observing points (black dots, centres of raster cells) within the visibility radius. Single viewsheds from the empty cells are not calculated. By A. Bucha Rášová.

[5] A similar approach was also used by S. Rana (2013) who demonstrated that total viewsheds can be reliably computed using only fundamental topographic features as targets.

The resulting viewsheds differ in two parameters: the visibility radius and the resolution of the input grid. As the basic radius, we have set 0.5 km and the cell size of 5 m; other models were produced for multiples of these parameters: 1 km/10 m, 2 km/20 m, 4 km/40 m, 8 km/80 m, 16 km/160 m and 32 km/320 m. The only exception is the model with a radius of 64 km, where we preserved the cell size of the preceding iteration (320 m). These parameters reflected the available processing capacity but still met the output requirements given that the details of the relief are not visible anyway at greater distances. As a consequence, over short distances, the models offer a relatively detailed raster reflecting the local relief; over long distances, representation of larger landscape features predominates. In the individual raster datasets, the values correspond to percentages of visible cells in the given radius (0–100, rounded up to the next integer value).

The total-viewshed datasets described above are referred to as TVIEWn, where "n" corresponds to the order given by the visibility radius (1 = 0.5 km, 2 = 1 km, 3 = 2 km … 8 = 64 km; Fig. 5).

3.2. Derived raster datasets

From the total viewsheds (TVIEWn), the 'differential' total viewsheds (DVIEWn) have been calculated, in which the values of the model lower by the order are subtracted. This adjustment is justified by the fact that models of lower order are always *de facto* included in the models with a larger visibility radius (Fig. 6). This way, the lower-order models contribute redundant information to the models of higher-order, especially if we want to find out which distance intervals were significant for visual contact. Since a model of the n-th order is always four times larger in area than the model of the (n–1)-th order, its values (percentages) represent four times larger areas of the landscape. Therefore, it is necessary to first normalise the values into the same scale (i.e., multiply the TVIEWn by four), subtract the model of the lower-order (TVIEW(n-1)) and then return the scale to the range 0–100 (i.e., divide values by three because subtracting the lower-order model reduces the possible range of values by a quarter). Therefore, the formula for the calculation of DVIEWn is as follows:

$$\mathrm{DVIEWn} = ((\mathrm{TVIEWn} * 4) - \mathrm{TVIEW}(n-1)) / 3,$$

where the values of individual TVIEW models correspond to the percentage of the visible territory within the given radius. When calculating DVIEW rasters, all input TVIEW rasters were first resampled to a uniform resolution of 8 × 8 m per cell using bilinear interpolation.

In the next step, we reduced the information contained in the eight DVIEWn models using multivariate analysis. In this way, on the one hand, it was intended to simplify the further data processing, on the other hand, this step pursued exploratory objectives: we were interested whether the values of different total viewsheds correlate with each other, to what extent and where. We used the PCA function in ESRI ArcGIS and all the DVIEWn rasters in their full extent as the input. Factor scores of the corresponding PCA components were transformed into separate raster datasets (PCA1–3), the values of which can be subjected to focal or zonal statistics. All PCAn values were calculated on a scale of 0–255 (unsigned 8-bit), and in this scale, they are displayed in Tables 3: A and 5 Possible interpretations of these components/gradients are detailed in the next chapters.

To find out whether the location of any archaeological sites reflects intentional human behaviour, it is first necessary to specify the overall character of affordances offered by the geographical area in which people were able to meet their needs. This task primarily means determining the relevant extent of the territory itself. In the case of our testing sample, we have defined the area for further calculations both by intuition (as an arbitrarily defined territory of 196 km2: PCAn-Reg layer) and as a summary buffer zone within a radius of 1 km around the barrow cemeteries (149.2 km2: PCAn-1km layer). Both alternative rasters were

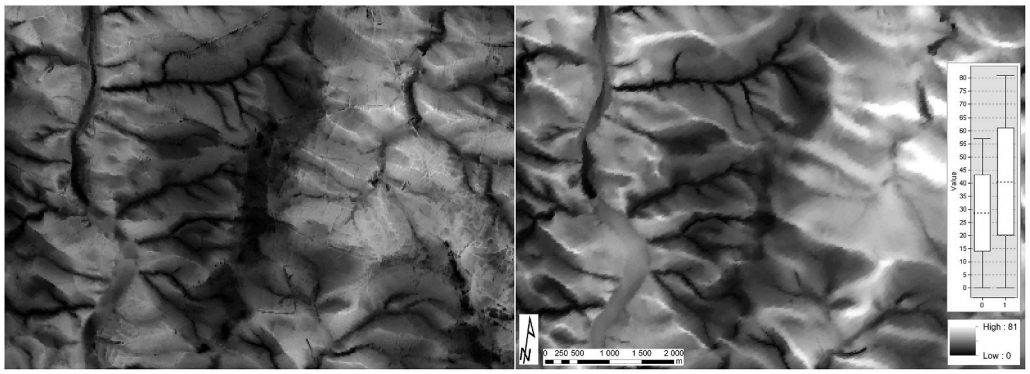


Fig. 4. Comparison of TVIEW2 (1 km observing radius) results with observer height set to 1.5 m and target height set to 0 m (left figure/box plot 0) and 1.5 m (right figure/box plot 1). The data in the left figure is partly affected by the fact that the modern features filtering step was skipped during the test calculations (for comparison purposes, however, the effect of this difference is negligible). By D. Novák.

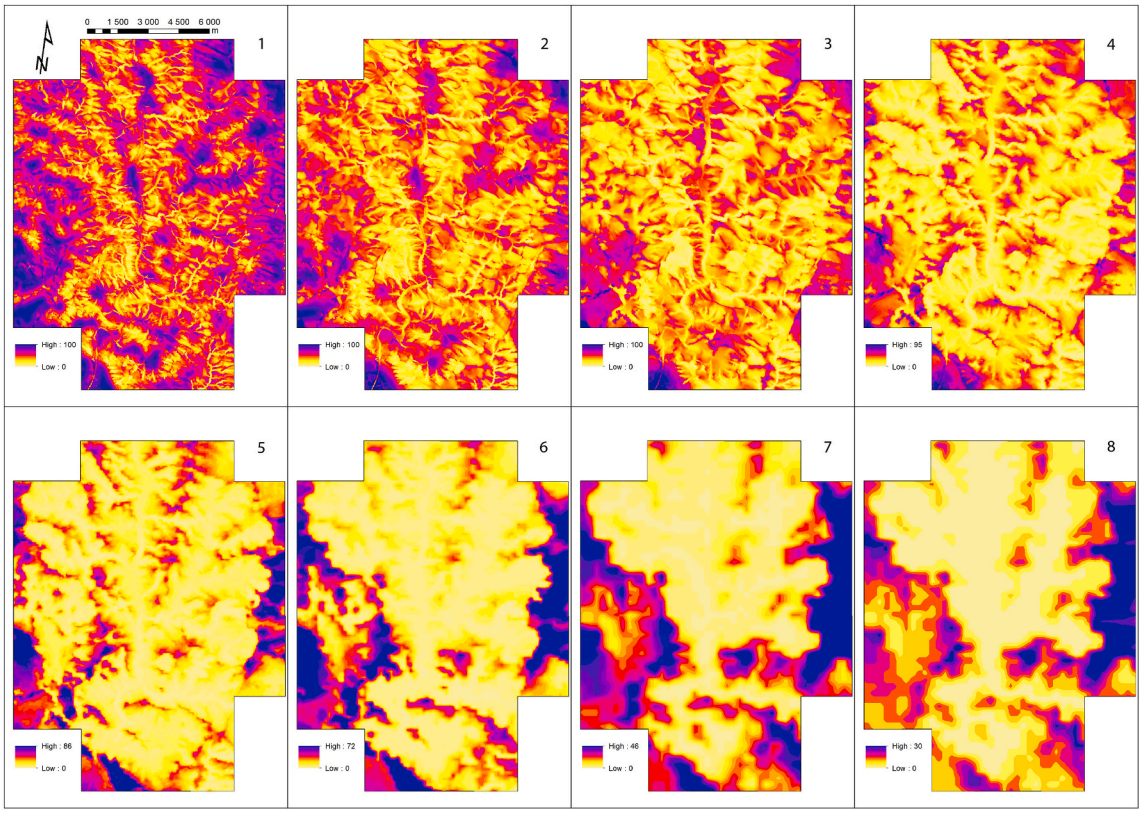


Fig. 5. TVIEW1-8 datasets for the study area. The completely different structure of visibility parameters for each visibility radius is obvious. By D. Novák. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

created by simply cropping the original PCAn layers in ArcGIS (cf. Fig. 8: A–B).

We consider the last-mentioned approach more suitable: an area defined by a radius of 1 km can be considered a realistic approximation of the Bronze and Iron Ages 'community areas' (in the sense of Neustupný 1998) in the region. Given the density of the barrow cemeteries in the landscape (thus also the density of assumed settlements), we may deduce that the areas beyond this distance were 'socially unusable' for the members of individual communities. The territories shared by one community may have been larger than this (and probably not concentric), but the distance between their main components (the settlement and the burial ground) was most likely less than 1 km (judging by the known cases from different periods of prehistory; cf. Dreslerová 1995; Neustupný ed. 1998). The more distant areas were probably both beyond the limits of efficiency of day-to-day activities and out of the immediate control of the community.

The visibility of any place in the landscape was determined by its relationship to the surrounding landscape, but this does not mean that the entire visible landscape was available for the same purpose as the place from which it is viewed. On the contrary, we believe that in prehistory, the landscape was generally quite strictly structured (Neustupný 1986; 1998) and the choice of a certain place for any activity was *de facto* considerably restricted. The builders of the burial mounds could take into account the lookout into the landscape over several kilometres, but the optimal location following such an evaluation had to be found within a narrow space available to burial activities of the given family or community. To identify the landscape potential at this scale – within the frame of local or relative affordances – we have calculated a 'relative

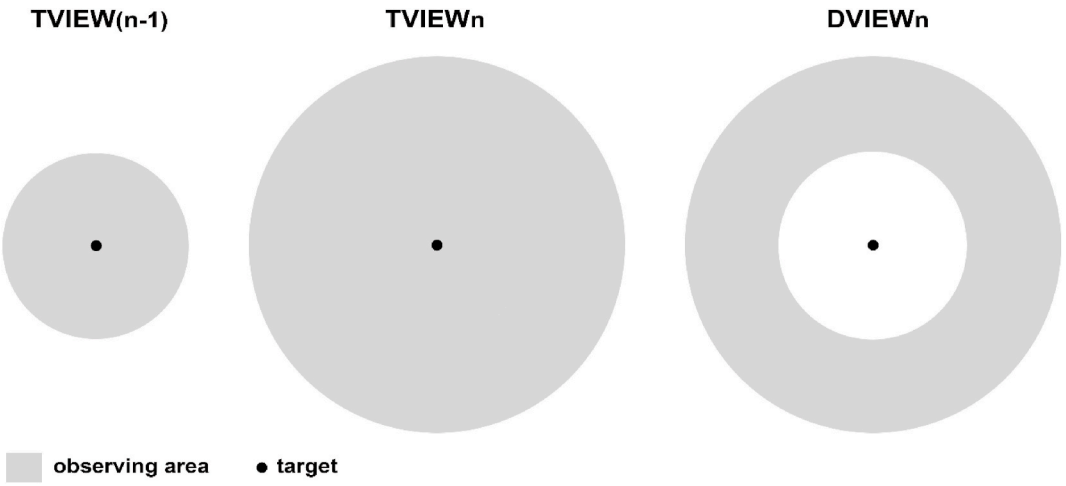


Fig. 6. Principle of the calculation of the DVIEW models.

total-viewshed gradient' (RPCAn) with relative values within a radius of 250 m (using the *Focal Statistics* function in ArcGIS, normalised by the standard deviation):

$$RPCAn = (PCAn - PCAn_mean) \,/\, PCAn_std,$$

where the 'mean' variable represents the mean PCAn value in a radius of 250 m and the 'std' variable is its standard deviation within this area.

3.3. Preparation of the archaeological data

As mentioned above, the test sample was selected as a set of barrow cemeteries in a well-surveyed region of southern Bohemia. The choice of this data was also related to the authors' previous experience with the chosen topic and region. Thus, it was possible to focus primarily on studying the properties of total viewsheds (which is the aim of this study), rather than paying attention to the complex archaeological interpretation of the data. The dataset was for the most part created in 2003 and based on published field surveys (Beneš et al. 1999). In the last two years, this data was also entered into the *Archaeological Map of the Czech Republic*[6] within the AIS CR research infrastructure. To a limited extent, we have verified the given data with our field survey in 2003, and in 2019–20, we added a few new sites found in the LiDAR images (Machová – John 2020) and by regional archaeologists (co-authors of this paper). The barrow cemeteries were digitised as polygons; solitary mounds received a buffer zone of a 45-m radius (6000 m2). This buffer radius was gradually reduced towards larger burial grounds so that the largest of them (max. at ca. 60,000 m2) gained no extension at all and were recorded with their actual size. This adjustment was intended to limit the influence of inaccuracies in the maps, the effects of incomplete archaeological knowledge and other secondary factors. For each cemetery, the number of visible (or known) mounds and their dating have been recorded (the latter, however, known only in a small number of cases).

3.4. Methods of statistical processing

Using the *Zonal Statistics* function in ESRI ArcGIS, individual site polygons were assigned various landscape properties including the values from the total-viewshed derivations (PCA1–3 and RPCA1–3). For further processing of this data, simple statistical methods were intentionally preferred since our data was new and sometimes difficult to understand: therefore, we tried to put the production of results under maximum control and to keep its parameters comprehensible and easy to modify.

In principle, we tried to find out whether the locations of the barrow cemeteries correspond to the natural frequency of the visual affordances of the landscape or not. If not, a certain degree of the intentionality of past human behaviour can be assumed. Most often, we used the nonparametric chi-square test, which compares the observed values (number of sites) to the 'expected' ones (i.e., numbers that correspond to the natural frequency of the given affordance). The advantage of this approach is that it shows not only the overall result but also the specific intervals of values that contribute most to it.

The PCA1–3 and RPCA1–3 values were divided into quartiles. We assume that if the location of cemeteries did not respect a specific landscape property (a visibility gradient in our case), the observed number of cemeteries should be divided evenly into quarters following the quartile values of the relevant variable, i.e., they should form four groups of about 30 sites (119/4 = 29.75). Relevant values for individual cemeteries were derived as mean values from the input rasters using the *Zonal Statistics* function. The probability of the observed distribution of the cemeteries being random was read from the common statistical tables (for quartiles, i.e., at three degrees of freedom).

4. Results

The distribution of values in our viewshed models is expressed in Fig. 7; this graph shows a smooth change in the composition of values in individual raster models resulting from the general structural characteristics of the Bohemian landscape. It is clear from the plots that as the observation radius increases, the usual proportion of visible surroundings decreases. The almost normal distribution for TVIEW1 (DVIEW1) changes to a significantly reduced visibility for TVIEW8 (DVIEW8).

The relationships of the models are well explained by the PCA (principal component analysis, see above). The results are quite clear (Table 3), which allows us to reduce the number of datasets in further analyses (PCA1–3). In particular, three basic axes of variability (factors/components) were identified, which can be understood as basic "gradients" of the visibility of the Czech landscape. Component 1 (PCA1 raster) defines the gradient of the short-to medium-distance visibility (500 m – 4 km). Component 2 (PCA2) is bipolar and shows the degree of contrast between the visibility over short distances (around 500 m) and medium distances (2–8 km). The positive pole of this factor means a high level of near visibility at the low level of the medium-distance visibility; the opposite is true for the negative pole. In other words, the factor scores in this case represent a measure of the 'distinctiveness' of a place

[6] For the data see https://digiarchiv.aiscr.cz/results?entity=lokalita&page=0&nazev=*mohyln%C3%ADk*:or.

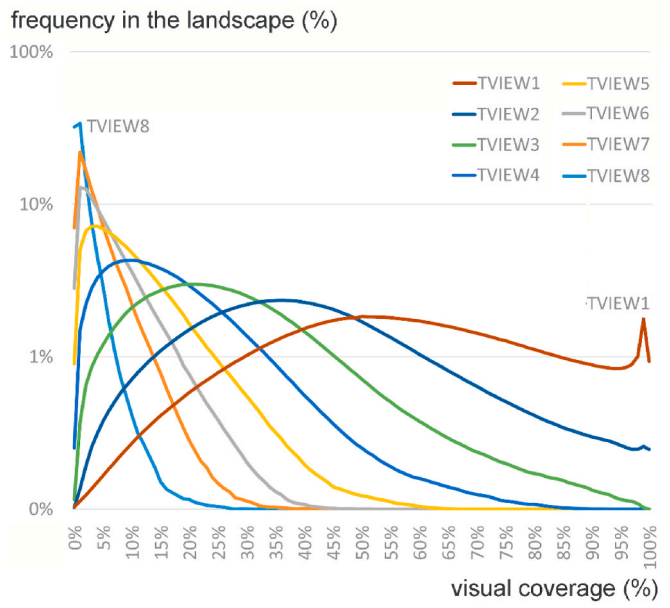


Fig. 7. Graph of the frequency of total visibility values in TVIEW1–8 models. Calculated for the whole territory of Bohemia. X-axis: values of total visibility (percentage); Y: frequency of the given values in the landscape, percentage on a logarithmic scale. The peak in the data for TVIEW1 (and partly also for TVIEW2) at the level of 99% is an artefact caused by the process of rounding the values to integers and has no informative value. By D. Novák. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

in terms of its visibility at a certain distance and at the same time its hiddenness from another distance. Component 3 (PCA 3) displays the visibility gradient over middle and long distances (8–16 or 4–32 km). Such results show that our total viewsheds are general enough to be reused in different applications and offer a significant explanatory potential to be analysed as a proxy to archaeological data and to interpret past human behaviour.

In order to test the validity of the viewshed models on a real example, the first task was to compare the visual properties of barrow cemeteries with the natural distribution of the viewshed/PCA values in the landscape under study. The data shows that for the first and second components (PCA1 and PCA2) the distribution of cemeteries is highly non-random; for PCA3, on the contrary, any non-randomness is improbable. We have obtained almost the same results using both options of the definition of the 'background' territory, i.e., both the 'region' and the 1-km buffer around the sites, which is preferred and displayed here (Table 4:A).

However statistically proven, the results for the PCA1 and PCA2 cannot be interpreted straightforwardly. The PCA1 shows a concentration of sites in the middle quartiles (Table 4:A), which means that the cemeteries avoided both too well visible and too hidden places. This is a result, which – given the interpretation of human behaviour – does not make much sense. A pitfall is also contained in the PCA2 results that demonstrate a clear preference of places well visible over the middle distance but hidden in the short view. Not even this result can be explained easily (cf. chapter 5).

A hint for the interpretation of both these data patterns is provided by the relative total viewsheds (RPCAn), which classify each point in the landscape according to the landscape properties in a radius of 250 m. The RPCA1 values offer an unequivocal result, i.e., a clear preference for areas with good short- and middle-distance visibility, while the RPCA2 data is statistically inconclusive (intentional 'hiding' of cemeteries in the short-distance view can therefore be excluded; Table 4:B). The RPCA3 gradient also does not point to any important relationship. The way we include the results in the overall interpretation of the data is discussed in the following chapter.

For individual PCA1–3 quartiles, mean values of other landscape properties were calculated, such as the mean altitude, mean relative elevation (above the minimum in a radius of 500 m) and the mean distance from watercourses creating valleys (cf. Table 5). Given that the total-viewshed values are always related to the relative elevation of the place (either in a positive or negative sense), we also made a similar calculation *vice versa*: we have determined the mean PCAn values for individual intervals (octiles) of the relative elevation raster (Table 6); these results will also be used in the discussion below.

Finally, the relevance of the total viewsheds has been tested with a simple predictive model. This was derived from two visual properties that show an almost obvious relation to intentional human behaviour: RPCA1 and PCA2 visibility gradients (Table 7; for the explanation see the next chapter). The frequency of barrow cemeteries in the prediction-model zones was also compared to their chronological classification, although this can be obtained only rarely and can play only an indicative role (Table 8).

5. Discussion

The PCA processing of total-visibility models of the whole territory of Bohemia has shown that, in general, the visibility of the earth's surface from various distances creates logical structures. These structures can be seen as an effect of the gradual increase or decrease in visibility of any place with changing distance of the observer. On a large scale, our analysis has captured three of these levels, which can be understood as 'gradients' of the general visual properties of the physical relief: (i) the gradient of short-distance and middle-distance visibility (PCA1); (ii) the gradient of the contrast between short-distance and middle-distance visibility (PCA2) and (iii) the gradient of middle- and long-distance visibility (PCA3; Table 3). We try to illustrate the meaning of these gradients on the example of a specific landscape section with a numerous and homogeneous set of archaeological sites – prehistoric barrow cemeteries (Fig. 8:B–C).

The studied territory (north from the municipality and castle of Hluboká nad Vltavou, South Bohemia) consists of a structured hilly landscape with the relatively deep Vltava valley crossing its centre from the south to the north (Fig. 8:A). The altitude of the territory varies over 140 m (354–593 m ASL). Apart from the altitude, the landscape can be described by its relative elevation (in our case, constructed as the difference between the actual altitude and the local minimum in a radius of 500 m) and the distance from larger watercourses (here defined by the catchment area of at least 1 km2). The horizontal and vertical distances from the watercourses (valley bottoms) are in this specific case closely related since the landscape as a whole is divided, apart from the central river valley, by a relatively regular grid of its smaller tributaries. The uneven but regularly articulated landscape relief allows reckoning with a relatively comprehensible visibility structure (Fig. 5).

The visibility gradient over short- and middle-distances (PCA 1) is clearly related to the altitude, the relative elevation and the distance from the valley bottom; it is, however, inversely proportional to all these variables (Table 5). At first, we were surprised by this fact because intuitively we supposed the opposite result, i.e., a positive correlation of the viewshed area and the elevation (dominance) above the surrounding landscape. This false expectation is connected with the fact that human perception (including ours) structures the landscape according to prominent (usually elevated) features, which it evaluates as logical and well visible entities, but does not measure analytically their quantitative surface values in the way a raster representation would do (which, on the contrary, does not identify the entities behind individual raster cells or their groups). From the analytical point of view, better visibility of lower-positioned sites is logical if we consider that lower-lying parts of the landscape (valleys) are concave in principle, and therefore, visible looking from above, while looking from below up to the watershed borders the number of convex shapes always grows and decreases the

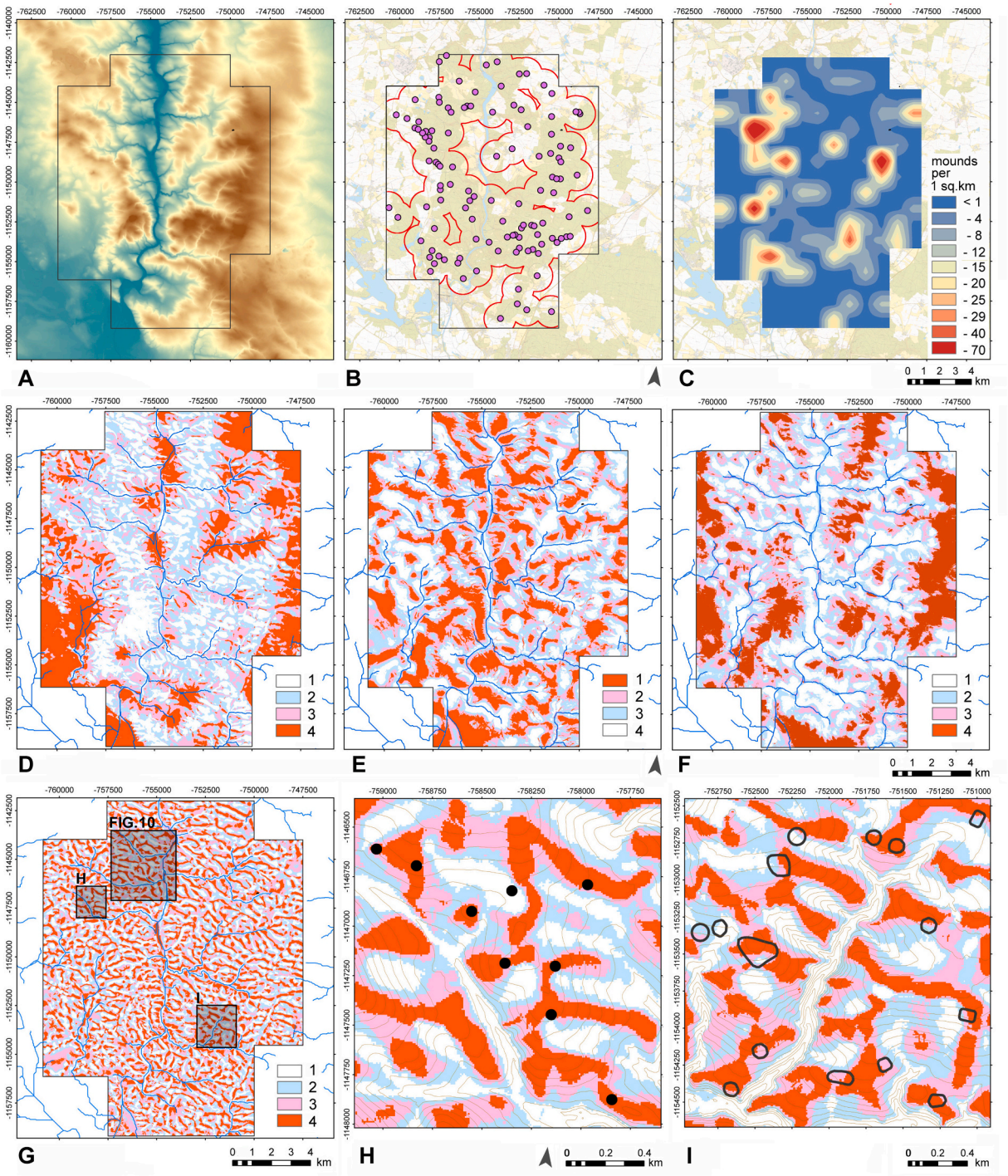


Fig. 8. Hluboká nad Vltavou, the area under study (PCAn-reg). A: DEM (min. 354 m, max. 593 m ASL); B: position of prehistoric barrow cemeteries (119 in total); red line marks the 1 km-buffer zone around the barrow cemeteries (PCAn-1km); C: density of barrows per 1 sq. km (max. 70); D: gradient of the PCA1 (split into quartiles); E: gradient of the PCA2 (quartiles marked on the inverted scale); F: gradient of the PCA3; G: model of the relative total visibility (RPCA1). Grey squares show the areas displayed in detail in Figs. 8:H, 8:I and 10; H: detail of the RPCA1 with the position of prehistoric barrow cemeteries (marked as points); I: ditto (cemeteries marked as polygons). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

visibility of the relief. This is also confirmed by Table 6, which unequivocally proves the fluent decrease of PCA1 values with increasing elevation. It also exemplifies that lower-lying (and better visible) terrain is visually more differentiated; thus it contains both well visible and more hidden viewpoints (the range of the PCA1 values, i.e., the difference between the minimum and maximum values, declines with growing elevation).

This approach to the PCA1 gradient explains the observed concentration of barrow cemeteries in the two middle quartiles of its values (Q2, Q3; Table 4:A): the PCA1 outer quartiles (Q1 and Q4), mostly avoided by the cemeteries, seem to characterise areas, which are situated either too close to the valley bottoms (i.e., in the area of the supposed settlements and economic activities; high PCA1 values) or too far from them (low PCA1 values). Most probably, therefore, the pattern of cemetery distribution is a consequence of the overall spatial structuring of the settlement areas, which is not based on the aspect of

Table 3
Results of the principal component analysis (PCA) of the DVIEW rasters for the territory of Bohemia. Factor loadings considered relevant are marked in bold. Components (factors) PCA1-3 are interpreted as the main visibility gradients in the landscape.

viewshed model	radius (m)	PCA1	PCA2	PCA3
DVIEW1	500	**0.86**	**0.47**	0.21
DVIEW2	1000	**0.93**	−0.08	−0.31
DVIEW3	2000	**0.75**	**−0.57**	−0.04
DVIEW4	4000	**0.51**	**−0.66**	0.44
DVIEW5	8000	0.29	**−0.52**	**0.57**
DVIEW6	16,000	0.11	−0.33	**0.50**
DVIEW7	32,000	0.01	−0.11	0.36
DVIEW8	64,000	−0.04	−0.07	0.19

visibility as such.

Nevertheless, good visibility over short and middle distances played an important role for the builders of the burial mounds. Given the above said, this may seem contradictory, but it is not: the short-to middle-distance visibility is not reflected in the positioning of the cemeteries in the landscape on the scale of its whole (the area from which the site was to be seen), but rather in the location of the cemetery in the context of its immediate surroundings (the area that was available to the community for use). This is documented by the distribution of the RPCA1 values, which shows that, within the area available for burials (with the possibility of choice within a radius of c. 250 m), places with optimal visibility were mostly chosen. The RPCA1 gradient clearly distinguishes between local convex and concave shapes, especially in the case of small ridges and gorges leading transversely to the main valley (Fig. 8:G–I). On this scale (in contrast to what we found for PCA1), it is valid that convex shapes are (over short and long distances) better visible than the concave ones in their immediate surroundings. On this level, a clear preference for convex shapes by the builders (Table 4:B) proves that within the potential burial areas (i.e., the socially defined space for a certain activity), good visibility of the burial mounds represented an important and intentional aspect.

The PCA2 gradient describes the importance of a place in terms of its middle-distance visibility (2–4 km) in contrast to short-distance viewshed (500 m): this way, it determines places either typical for one or the other type of visibility (contrary to PCA1, where short- and middle-distance viewsheds correlate). PCA2 values may seem to correlate negatively with PCA1 values; from the mathematical point of view, however, both factors are independent. Table 5 shows that the negative pole of the PCA2 gradient is connected with higher viewpoints located further away from valleys, the positive pole with the opposite (lower areas, close to valley bottoms). Essentially, the same is attested in Table 6, where the mean PCA2 gradient decreases from locations close to the valley bottom (positive factor pole) to places with larger vertical (and, logically, also horizontal) distance from them (negative PCA2 pole factor).

The PCA2 may seem to best reflect the intentions of the mound builders, especially for the significant concentration of the cemeteries in the lowest quartile of values. Low PCA2 values (cf. Tables 3 and 4:A) are best suited to describe higher viewpoints in the upper parts of the hills and ridges (Fig. 8:E). These are places with good middle-distance

Table 5
Mean values of the altitude, elevation (above minimum within the 500 m radius) and distance from water streams, calculated for individual quartiles (Q1–4) of the visibility gradients PCA1–3.

PCAn	quartile	mean altitude (m ASL)	mean elevation (m)	mean distance to stream (m)	
PCA1	**Q1**	465	41	549	
PCA1	**Q2**	469	41	544	
PCA1	**Q3**	461	31	486	
PCA1	**Q4**	449	19	475	best on short/ middle distance
PCA2	**Q1**	469	48	605	best on middle distance/worst on short distance
PCA2	**Q2**	462	35	552	
PCA2	**Q3**	459	28	499	
PCA2	**Q4**	454	20	392	worst on middle distance/best on short distance
PCA3	**Q1**	445	35	299	
PCA3	**Q2**	451	31	394	
PCA3	**Q3**	466	33	568	
PCA3	**Q4**	488	33	857	best on long distance

Table 6
Values of the visibility gradients (PCA1–3), calculated for the octiles of the relative elevation (above the local minimum in 500 m radius).

relative elevation (octiles of values)		PCA1		PCA2		PCA3	
	area (sq. km)	range	mean	range	mean	range	mean
0–8.3 m	138.1	156	89	121	135	88	66
8.3–17.8 m	327.4	139	75	109	137	99	69
17.8–27.9 m	439	149	67	109	130	105	71
27.9–39.1 m	369	127	57	109	127	93	68
39.1–50.9 m	339.3	137	54	117	121	98	68
50.9–67.7 m	231.7	127	52	131	114	99	68
67.7–88.2 m	88.2	97	50	131	108	104	70
88.2–150.9 m	26.6	80	48	82	103	85	66

Table 4
Distribution of barrow cemeteries in quartiles of the PCAn and RPCAn values, together with the results of the chi-square test. The quartile boundaries are given on the scale of 0–255 (part A) or multiples of the standard deviations (part B). The critical value at the 99% probability level is 11.35 at three degrees of freedom. Statistically relevant chi-square values are highlighted in bold, as are the quartile numbers of sites that most affect them.

quartile	*quartile value*	*no. of cemeteries*	*quartile value*	*no. of cemeteries*	*quartile value*	*no. of cemeteries*
Part A	**PCA1**		**PCA2**		**PCA3**	
Q-1	46	15	117	**46**	60	30
Q-2	59	**33**	127	29	67	30
Q-3	72	**49**	137	21	75	29
Q-4	140 (max)	22	170	23	126	30
chi-quadrat		**22.14**		**13.00**		0.03
Part B	**RPCA1**		**RPCA2**		**RPCA3**	
Q-1	−0.614	13	−0.392	38	−0.491	34
Q-2	0.046	23	0.104	28	0.007	32
Q-3	0.646	32	0.544	29	0.506	29
Q-4	5.63 (max)	**51**	5.544	24	8.482	24
chi-quadrat		**32.58**		3.30		2.09

Table 7
Numbers and density of barrow cemeteries in the main zones of the predictive model, calculated from visibility gradients PCA2 and RPCA1.

probability scores	area (sq. km)	no. of cemeteries	no. of mounds	cemeteries per 1 sq. km	mounds per 1 sq. km	mounds per cemetery
low (1–3)	57	15	100	0.3	1.8	6.7
middle (4–9)	102.5	62	508	0.6	5.0	8.2
high (12, 16)	36.4	42	431	1.2	11.8	10.3
total	196	119	1039			

Table 8
Dating of the barrow cemeteries belonging to individual zones of the predictive model.

probability score	Early Bronze Age (later Únětice & Věteřov c.; 1800–1600 BCE)	Middle Bronze Age (Tumulus c.; 1600–1300 BCE)	Late/Final Bronze Age (Urnfield c.; 1300–800 BCE)	Hallstatt period (800–550 BCE)
low (1–3)	-	-	-	5
middle (4–9)	4	9	5	7
high (12, 16)	6	6	2	4

distance visibility, but situated in such a way that its increase correlates with a decrease of short-distance visibility: very often, therefore, these are places, which from below locations stay hidden behind a convex terrain edge or slope curvature.

Here as well, it is necessary to approach the results cautiously. If looking for an original human intention at this point, we would have to assume that the human intention was not only to guarantee good medium/long-distance visibility of the place but also its hiddenness over the short distance. However, it is difficult to find an adequate explanation for such behaviour, which, anyway, does not seem very probable since the distribution of the cemeteries according to the locally available values (RPCA2) looks perfectly random (Table 4:B).

We can explain the opposition of PCA2 v. RPCA2 by the fact that, on the larger scale, constructing burial mounds on elevated places above valleys was not to increase their visibility, but it was a consequence of the general properties of the landscape and its overall structuring within the borders of settlement areas (community areas, cf. Neustupný 1998). The choice of elevated parts of the terrain for burials was not necessarily connected with their visibility. E.g., the valley bottoms and lower slope parts could have been reserved for residential and production areas; the overall spatial patterning of activities within the community areas could have played a role as well. The symbolic meaning of higher-elevated places could have also been considered (for the symbolic meaning of verticality in prehistory in general see Neustupný 1997), especially if it was associated with the ancestral world. From this point of view, a certain 'hiddenness' of these places was, on the general level, rather an unintentional, but tolerated property of the burial areas, and it did not by itself play a role in the choice of their location.

The situation may also explain why the PCA2 gradient appears as an important factor for the distribution of prehistoric barrows, but it does not create any clear 'regularity': 79 barrow cemeteries fall within the two lower PCA2 quartiles (Q1-2), 40 belong to the upper ones (Q3-4); so about a third of cases do not correspond to the given rule. In our opinion, this is a consequence of the fact that the PCA2 gradient was only one of the factors, and perhaps not even the main one, in deciding for a specific location of the burial areas. We may even assume that, if possible, other locations were considered as well. It seems that here we face a general aspect of human behaviour (but an uneasy problem for research), i.e., the impossibility of reducing it to a single cause or factor.

The complexity of this question can be illustrated by the relationship between the total visibility and the natural landforms (as recently calculated for the whole of Bohemia by Novák 2018; cf. also Kuna 2006). Projecting the barrow cemeteries on the terrain (landform) classification model (TCM; Novák 2016), we can see that the concave shapes (marked 1–6 here, mainly valleys and lower parts of slopes) are significantly less represented in burial grounds than in the overall landscape: it can be, therefore, assumed that these landforms were mostly avoided (Fig. 9:A). Moderately elevated landforms (7–12) are represented in proportion to their occurrence in the landscape – this is why it is not possible to prove any intentionality of their use. There is a clear preference for only two of the landforms – local hills (13) and hilltops (14). We can see, however, that the mean total visibility of these landforms (compared to the others) increases with the increasing viewshed radius and the above-average values are achieved only from a radius of 2000 m upwards (Fig. 9:B). This is in line with our above conclusions and it confirms our assumption that the above-standard visibility at medium and long distances (with poorer short-distance visibility) was not the main aspect of the burial-ground locations. It is conceivable that the potential meaning of hills and higher plateaus lay in the elevation itself (being, e.g., a product of the respect for ancestors), while the good visibility from medium and larger distances was rather an indirect consequence of this choice. From this point of view, however, any absolute measure of the landscape cannot be completely reliable, because the decisive factor was a relationship – the relative position (in this case expressed by the difference in elevation) between the world of the living and the dead. To a large extent, the core of the main problem of visibility studies can be illustrated here: a well-visible 'place' (a hill with mounds) can be observed from various angles or distances and firmly attached to a certain meaning (the world of the ancestors), even if the object that is

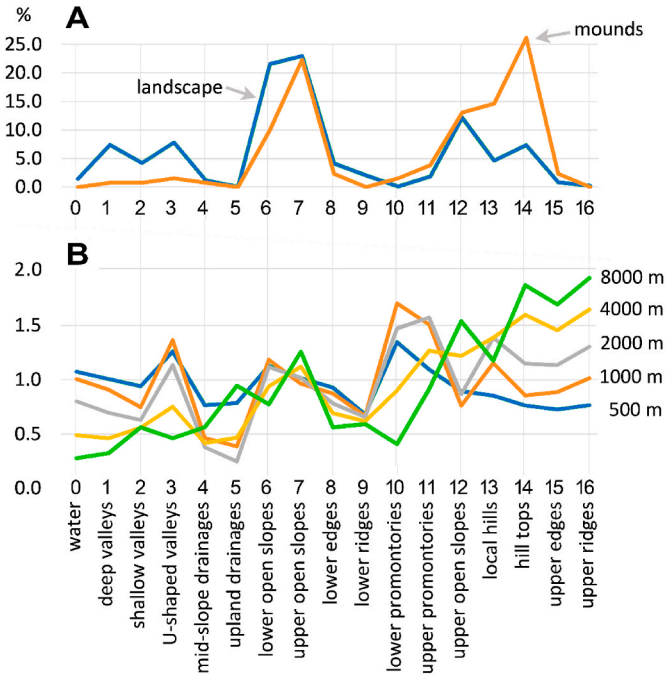


Fig. 9. Graph of the terrain classification model values. A. Relationship between the area of TCM landforms in the given region (blue) and prehistoric barrow cemeteries (ochre). B. Mean total visibility of individual TCM landforms in the region under study. Values from TVIEW1-5 are shown. TCM landforms are described by numbers (cf. Novák 2018) and names (B). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

responsible for that meaning (burial mounds) is hidden.

The PCA3 total-viewshed gradient describes the visibility over long distances. Its values grow with the altitude of the territory and the distance from the valley bottoms, but independently from the local elevation. This is a logical consequence of the landscape character, the visibility of which over long distances reflects mainly major landforms, but much less their local elevation context. The general landscape character in the geographic area in question does not allow for long-distance visibility (c. 8 km and more; Fig. 5) and if so, it concerns its border parts – slopes directed towards the lower and flatter landscape in its surroundings. This is not surprising because, given the distribution of the barrow cemeteries, the PCA3 visibility gradient is unimportant (Table 4), although, theoretically, this could be partly caused by the fact that the border zones, visually opened outwards to the surrounding landscape, have been mostly deforested in the past and the barrow cemeteries may not have been preserved in them.

We can summarise that, for the understanding of the location of prehistoric barrow cemeteries, the PCA2 and RPCA1 gradients have proved as the most important (if not the only) visual affordances reflected in human behaviour in the Bronze Age. If we assign the quartiles of their values a number according to the importance they had for the location of the cemeteries (i.e., in the case of RPCA1 from 1 to 4, in the case of PCA2 from 4 to 1), we can use them to create a simple predictive model. Multiplying these layers, we obtain a raster dataset with nine values (multiples form values 1, 2, 3, 4, 6, 8, 9, 12 and 16), which summarise the probability of the occurrence of barrow cemeteries in various parts of the landscape (Fig. 10). In Table 7, we see that the resulting raster has a certain predictive potential: scores 1–3 correspond to a low density of barrow cemeteries (0.3 in 1 km2 on the average); scores 4–9 to middle density (0.6 in 1 km2) and scores 12 and 16 to a high density (1.2 cemeteries in 1 km2). Something similar comes out even if we consider the number of mounds (from 1.8 to 11.8 barrows/km2) instead of the number of cemeteries. To a certain degree, also the average number of mounds per cemetery (6.7–10.3) grows with the probability score.

These results represent a clear trend, but not a strong regularity. It seems that, if we had a better idea of the chronology of the barrow cemeteries, we would be better able to capture their spatial preferences. The behaviour of barrow builders may have changed over time, as they come from a period of 1600 years. Although we cannot draw reliable conclusions from the known dating of the barrows because the number of dated finds is small, it seems that visibility as a factor in the barrow location was more important in the Bronze Age, especially at its beginning (Table 8). Since we do not know of any earlier settlement in the given region (before the Bronze Age), the positioning of the barrows in well visible places can also be understood as part of a starting settlement strategy, i.e., to delimit and to mark space by individual communities in the course of the colonisation of the territory. On the other hand, a large part of the predictive-model values is formed by the medium-distance visibility contrasting to near visibility. If we want to attribute some intention to the human behaviour behind this phenomenon, we would have to look for it in the concept of verticality rather than visibility as such. It can then be assumed that at the beginning of the Bronze Age a certain (vertical) landscape structure was created, which had a deeper meaning and divided the landscape into the (lower) world of living communities from the (higher) world of ancestors. In a further development, this structure loosened as other factors came into play, such as social differentiation within the communities and separation of the elites from other community members in burial practices (Chytráček et al. 2021).

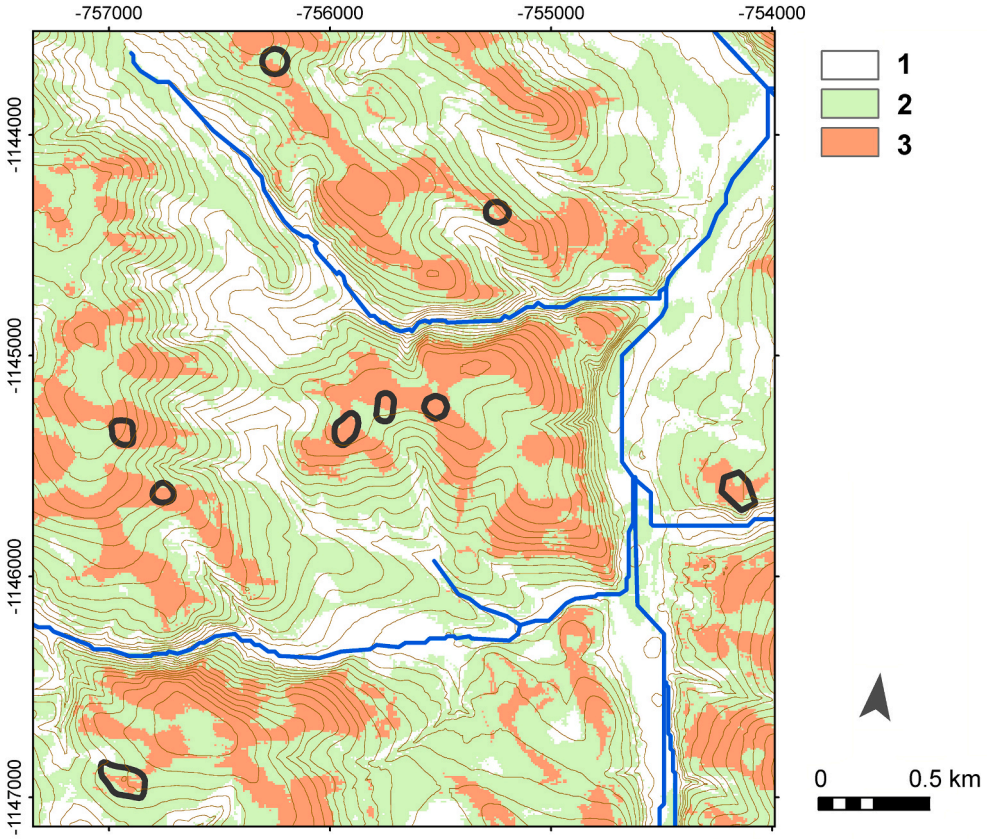


Fig. 10. Predictive model calculated from the gradients PCA2 and RPCA1. 1: low probability of occurrence of prehistoric barrow cemeteries (values 1–3); 2: middle probability (values 4–9); 3: high probability values (12 and 16). The original values were obtained as multiples of the values of the given data rasters, reclassified to values 1–4 (cf. chapter 5).

6. Conclusions

In conclusion, it can be summarised that the calculation of total viewsheds for larger territories has proven possible. However, high computational requirements may make these calculations unavailable to many users. Therefore, we have computed eight total viewsheds with various parameters for the whole territory of Bohemia and provided them for free use. We also pointed out the specific methods which can be used to reduce the computational complexity of producing total viewsheds.

We have constructed both total viewsheds (TVIEWn) and 'differential' viewsheds (DVIEWn) which in each model exclude the values of the lower-order models (i.e., models with smaller viewshed radius). In the case of our series of viewsheds, the lower-order models always cover a quarter of the territory of the next higher-order viewshed, which brings an unnecessary, though not essential, redundancy of data.

The mutual comparison of eight total viewsheds (DVIEWn) of Bohemia has demonstrated a certain degree of general logic of the visual properties of the earth's surface, which can be expressed with three gradients: the short- and middle-distance visibility gradient (PCA1, 500 m to 4 km); the gradient of the contrast (opposition) between the short (500 m) and the middle distances (2–8 km; PCA2) and the visibility gradient from the middle and long distances (PCA3: 8–16 km, resp. 4–32 km). We have studied the importance of these gradients on a selected part of South Bohemia.

This example has demonstrated some relations between the total-viewshed values and the occurrence of prehistoric barrow cemeteries (Bronze and Hallstatt Ages, c. 2000–400 BCE). These relations are neither simple nor straightforwardly explicable. The results obtained to date show that the frequently observable choice of elevated areas above the valleys not necessarily has to be primarily explained by their good visibility from middle distances, even though statistics may evoke this sort of conclusion. It is more probable that good visibility of these landscape parts from the middle distance was a by-product of the spatial behaviour primarily determined by other factors (mainly by the overall structure of the community areas, symbolism of verticality, etc.). If people followed these factors, they also had to tolerate the fact that, when viewed from below, i.e., from the probable location of settlements and production areas (in the short-distance view), the burial grounds were often hidden behind the horizon.

Nevertheless, good visibility (over short and middle distances) was still important when choosing a place to establish a burial monument, measured, of course, in the context of local affordances (in a 250-m radius), not of the whole region.

This conclusion leads us to emphasise the contextual nature of any geographic affordance in the past landscape. The observed patterns in the distribution of archaeological sites cannot be explained by themselves, but only in the context of an area that was available to the community and, for practical and social reasons, constrained in many respects. The given 'affordances' consist of multiple spatial levels (scales) and they are not the result of only the physical properties of a certain territory, but also of social, economic, symbolic and historical factors. These factors can also have the effect that the relation of human behaviour to the landscape is expressed on each spatial level (scale) somewhat differently.

We would also like to point out that there is a fundamental difference between the 'visibility of the terrain' and the 'visibility of a place' (understanding of the landscape). The discrepancy between these two aspects can be illustrated by the relationship between the visual prominence of a place (intuitively understood as a well visible entity) and its real visibility in terms of raster-map values. As an example from another part of the Bohemian landscape let us mention the legendary Říp mountain in northern Central Bohemia, which can be seen practically from the whole surroundings in a radius of tens of kilometres and represents one of the most visually significant places in the whole country. However, in the total-viewsheds, the values of its surface lie mostly below 50%. The raster map, contrary to human perception, does not 'know' that various sides of the mountain belong to the same landscape entity, and therefore it does not connect them, at least not without further calculations. In a way, this finding is also relevant for the assessment of places where burial mounds were established. The significance of these places need not have been derived from the possibility to observe (whether all or at least some) individual barrows, but it may have sufficed to recognize the higher entity, a major landform (peak, ridge, etc.), of which people knew it was connected with the given purpose and its specific social meanings.

Visibility of the earth's surface over a certain distance depends on many additional factors, such as the individual abilities of the observer, the size of the target and its visual context (relation to the horizon, vegetation cover etc.; Higuchi 1983; Ogburn 2006). As we have tried to suggest, there is one more point with the issue of visibility: the difference between the 'visibility' of a surface or landform and the 'recognition' of it as a meaningful entity. However, the latter is no longer just a question of sensory perception, but rather of cognitive evaluation, to which raster maps can provide the necessary basis, but not a complete key. Full exploitation of the potential of total viewsheds opens, therefore, another and in our view essential subject, which is the transformation of quantitative raster data into reasonable qualitative entities that are appropriate to human thinking and behaviour.

Attempts to understand the principles of terrain visibility have brought us to one conclusion, which may seem disappointing, but it is not. Through our research, we realised that the landscape could have been structured according to meanings that no longer exist in today's world, in this case, the verticality that distinguished the fundamental parts of the human world (the landscape of the living and the landscape of the ancestors). Because there is no such distinction in the landscape today, archaeologists have so far offered other solutions, more rational from the modern point of view. Among them, the practical or social impact of terrain visibility played a role as one of the main options. It is quite possible that we were wrong at this point (as in other cases where we encountered different ontologies), but we can only come to this conclusion after examining the given interpretive paths. We believe that a systematic study of total-visibility models without their a priori rationalisations can bring further surprises, and it is therefore very important to continue working on this topic.

Credit roles

Martin Kuna: conceptualisation, Methodology, Investigation, writing, Project administration. David Novák: conceptualisation, Methodology, writing. Alexandra Bucha Rášová: Methodology, writing. Blažej Bucha: Methodology. Barbora Machová: Resources. Jiří Havlice: Resources. Jan John: Resources. Ondřej Chvojka: Resources.

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